

FROM PROMPT TO PRODUCTION

Build, evaluate, observe and optimise a hosted agent

One agent. One use case. One improvement loop.

AI GENIUS • SEASON 6 • EPISODE 2

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FROM PROMPT
TO PRODUCTION

Build, evaluate, observe and
optimise a hosted agent

One agent. One use case.
One improvement loop.

ONE QUESTION DRIVES EVERYTHING

"Can I raise a \$6,200 purchase order
for integration licences this week?."



POLICY
Written rules
and thresholds



DECISION
Temporary CFO
exception



APPROVAL
The right person,
right process

THE POLICY ANSWER LOOKS RIGHT



AGENT ANSWER (POLICY ONLY)

This purchase exceeds the \$5,000
threshold. You need two written
quotes and Category Manager
approval before raising the PO.

Source: [Procurement Policy](#)



POLICY
DOCUMENTS

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DEMO 1: ASK THE AGENT

Same question. Policy only.



Procurement Assistant

Can I raise a \$6,200 purchase order for
integration licences this week?



This purchase exceeds the \$5,000 threshold.
You need two written quotes and Category
Manager approval before raising the PO.

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POLICY: A WHOLE
OPERATING PICTURE

THE MICROSOFT IQ LANDSCAPE

WHY? AS

Family documents

Recent decisions in email
and Teams that change
how policy is applied

FOUNDRY IQ

Mail, chats
+ files

WORK IQ

Mail, chats
+ files

FABRIC IQ

Data + semantic
context

WEB IQ

External
knowledge

Today we focus on Foundry IQ.

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RUBRICS DEFINE "GOOD" FOR THIS JOB

DEMO 3: EVALUATE THE BEHAVIOUR

Open the prepared dataset and evaluation run.

GENERIC EVALUATORS

Groundedness

Relevance

Coherence

Fluency

RUBRIC SCORE
(Policy Answer)

0.38

Poor

Same answer. Different definition of "good".

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THE DATASET IS THE SPECIFICATION



Exception
applies

Software licence
in migration scope



Exception does
not apply

Hardware
Normal route



Exception
expired

After 14 Nov
Normal route



High value
purchase

Over \$10k
Needs 2nd approver



Ambiguous
inputs

Ask for missing
information

Expected outcomes teach the agent what right looks like.

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THE NEXT TEST CASE IS HIDING IN A TRACE



User



Agent



Policy
retrieval



(No workplace
context call)



Answer
(Wrong route)



Tracer. show us what the agent actually did.

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AGENT OPTIMISER CLOSSES THE GAP

BASELINE

Instructions

Use policy to answer
procurement questions.
Ask if info is missing.

Tool description

search_workplace_context
"Search recent workplace
messages."



WINNING CANDIDATE

Instructions

Check for recent decisions
that may change or override
formal policy. Use them if
current and in scope.

Tool description

search_workplace_context
"Find recent approvals, exceptions
or changes that affect policy."

Rubric score 0.38 ➡ 0.86 +126%

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DEMO 4: INSPECT THE WINNING CANDIDATE

Compare prompts, tools and results.

PROMPT DIFF

- Use policy to answer...

- Ask if info is missing.

- Check for recent decisions

that may change policy.

- Use if current & in scope.

- Cite decision source.

TOOL DIFF

- "Search recent

workplace messages."

- "Find approvals,

exceptions or

changes that affect

policy and approvals."

RUBRIC SCORE

0.38

➡

0.86

Better instructions. Better tool guidance. Better outcomes.

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FROM EXPERIMENT TO ENGINEERING



Commit



Deploy
agent



Run
evaluations



Rubric
gate



Publish
results



Promote
(optional)

```
rubric_score=$(foundry eval summary --run run_123 --metric rubric)
threshold=0.38
```

```
if (( $(echo "$rubric_score < $threshold" | bc -l) )); then
```

```
echo "Rubric gate failed: $rubric_score < $threshold"
```

```
exit 1
```



PASSED

Repeat. Measure keep improvements safe.

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Q&A

OPTIMISE

GROUND

OBSERVE

EVALUATE

Building an agent is easy.

Building one that stays useful is an engineering discipline.



The slides orient. The demos do the teaching.

Can I raise this purchase order?

“Can I raise a \$6,200 purchase order for integration licences this week?”



POLICY

Written rules



DECISION

Current exception



APPROVAL

Right route

Same question. Different answers.

DEMO 1

Ask the agent

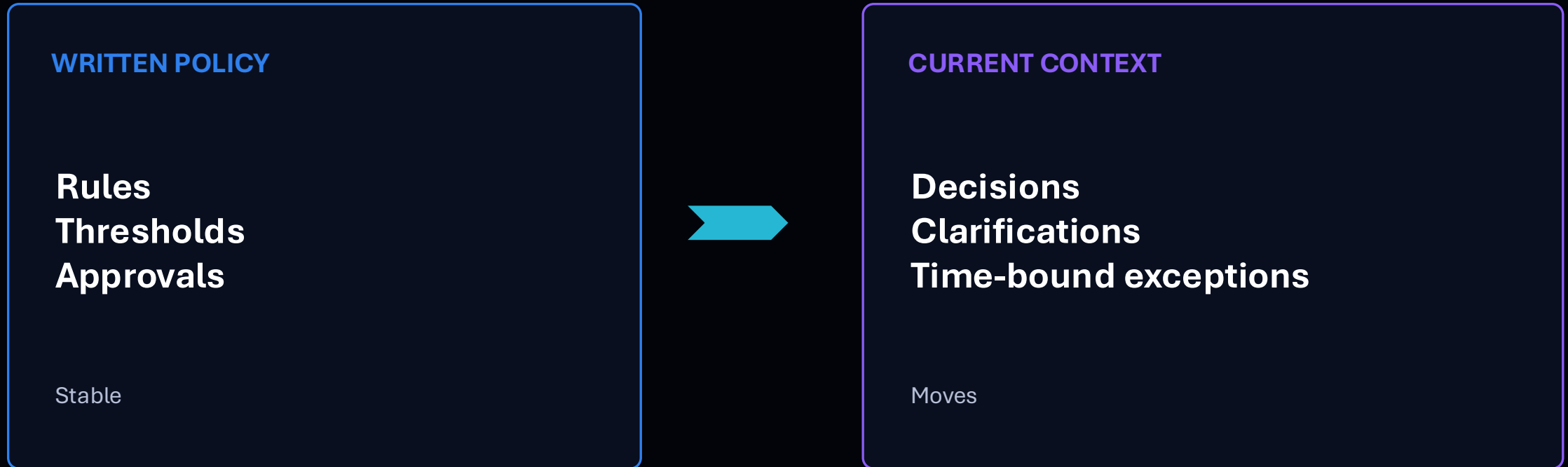
Same agent. Same question. Policy only.



**The policy answer
looks right.**

Confident. Grounded. Incomplete.

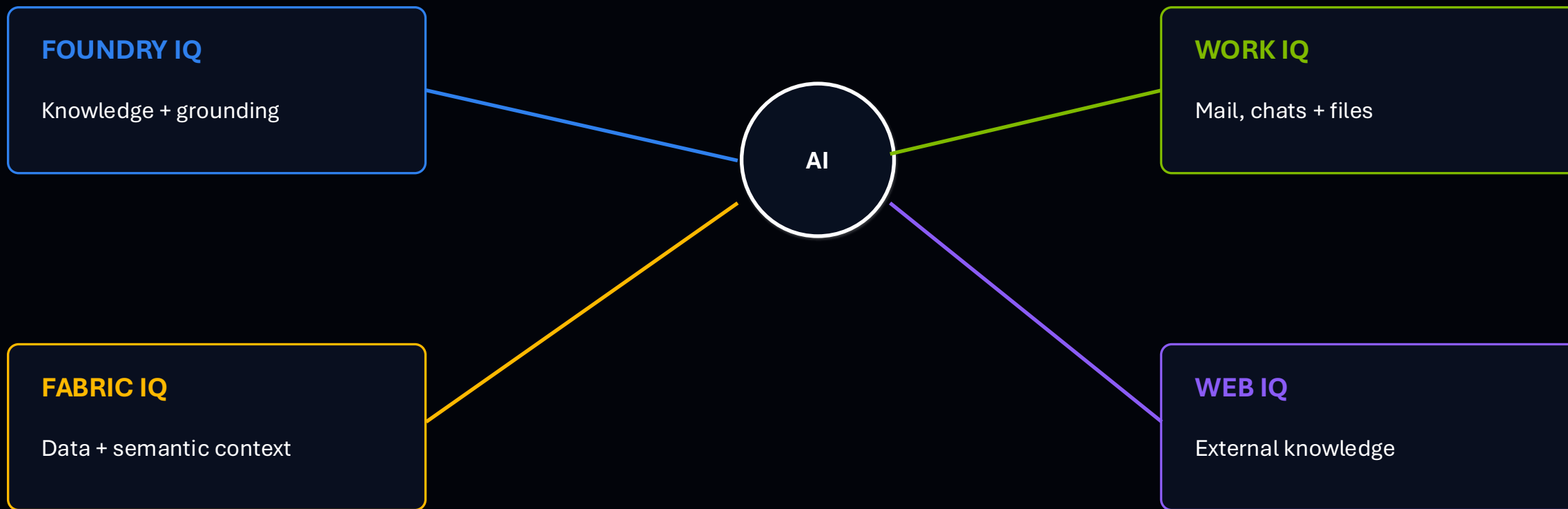
Policy is not the whole operating picture.



The agent needs both.

The Microsoft IQ landscape

One slide. Then back to the agent.



Today: Foundry IQ plus a hosted-agent function tool.

DEMO 2

Ask the identical question again.

POLICY ANSWER

**Two quotes +
Category Manager
approval**

Policy only



CONTEXT-AWARE ANSWER

**Interim ERP route
Single approver
Expires 14 Nov**

Policy + current decision

More context changed the route.

Both answers can score well.

GENERIC QUALITY

- ✓ Grounded
- ✓ Relevant
- ✓ Coherent
- ✓ Fluent

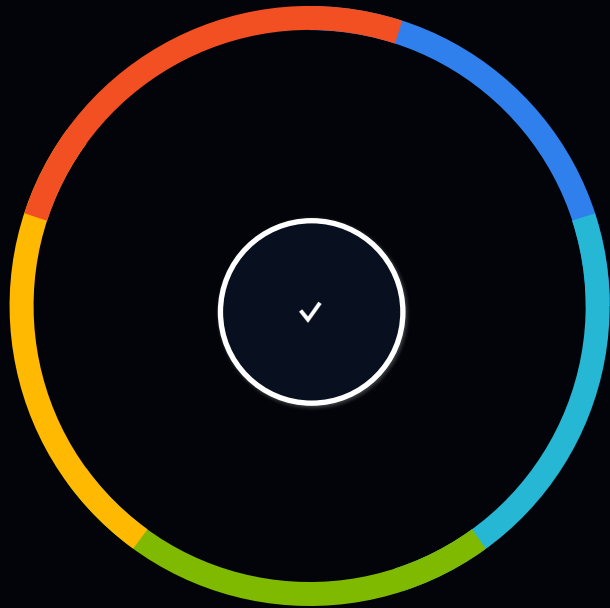
≠

BUSINESS SUCCESS

- ✗ Right exception
- ✗ Right approver
- ✗ Right dates
- ✗ Right action

Generic evaluations can be correct and still incomplete.

Define what “good” means for this job.



Policy + exception accuracy	25%
Current context use	25%
Decision awareness	25%
Approval-path correctness	15%
Actionability + transparency	10%

Business-specific rubric = primary signal.

DEMO 3

Evaluate the behaviour.

PREPARED EVALUATION RUN

GENERIC EVALUATORS



RUBRIC SCORE

0.38

Needs improvement

Same response. Different definition of “good”.

Expected outcomes are the specification.

Positive, negative and boundary cases stop the optimiser learning the wrong lesson.



APPLIES

**Licence
In scope**



DOES NOT APPLY

**Hardware
Normal route**



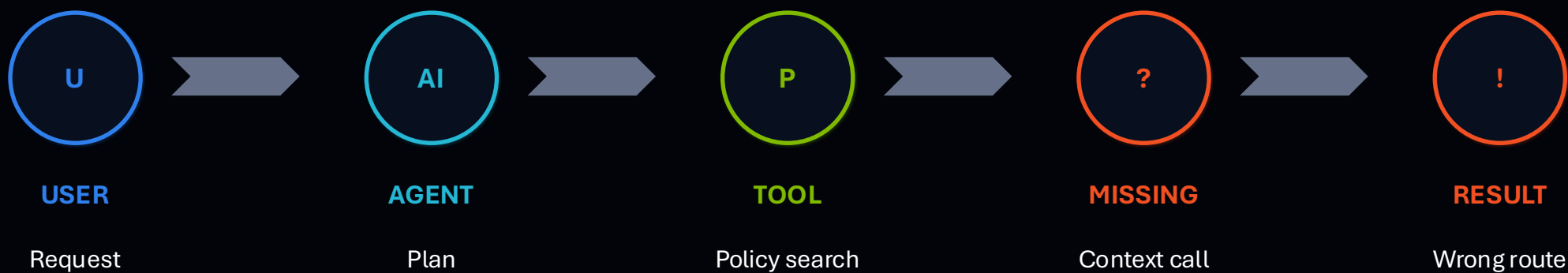
EXPIRED

**After cutover
Normal route**

The optimiser learns from outcomes, not assumptions.

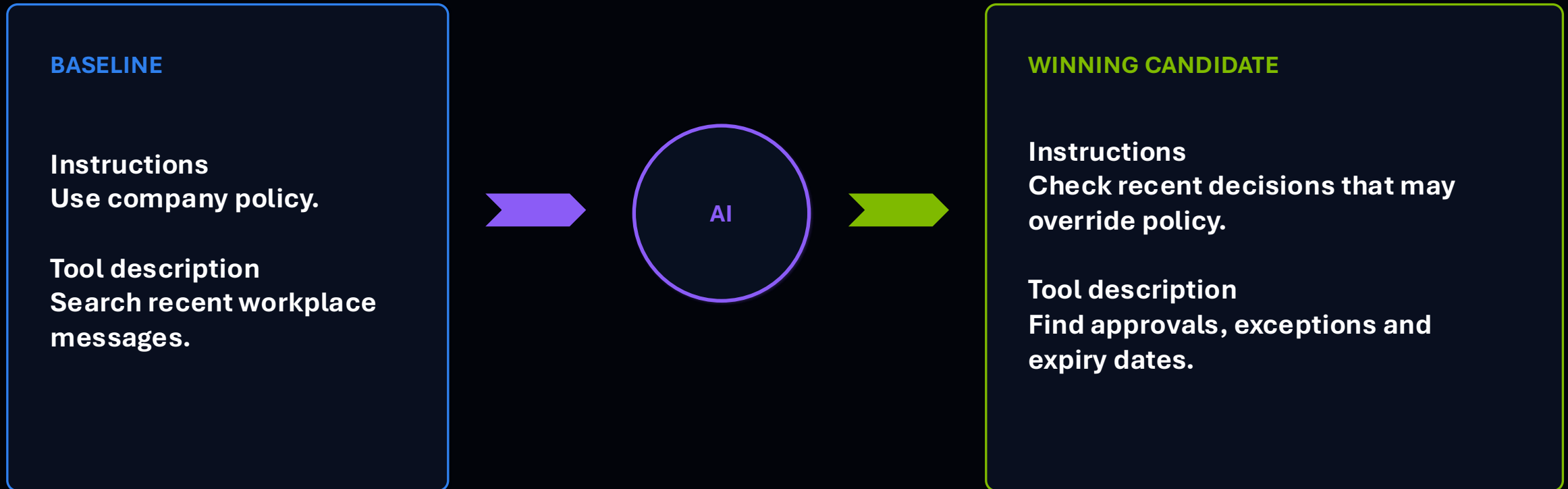
The next test case is hiding in a trace.

OpenTelemetry shows what the hosted agent called, skipped and returned.



Trace → dataset → regression test

Measure. Improve. Compare.



0.38 → 0.86

DEMO 4

Inspect the winning candidate.

Open the completed optimisation run.

BEFORE

Vague instructions
Weak tool guidance
Context sometimes skipped



AFTER

Explicit recency checks
Clear tool purpose
Scope + expiry handled

Same dataset. Better candidate.

Every change is measured. Every release is earned.

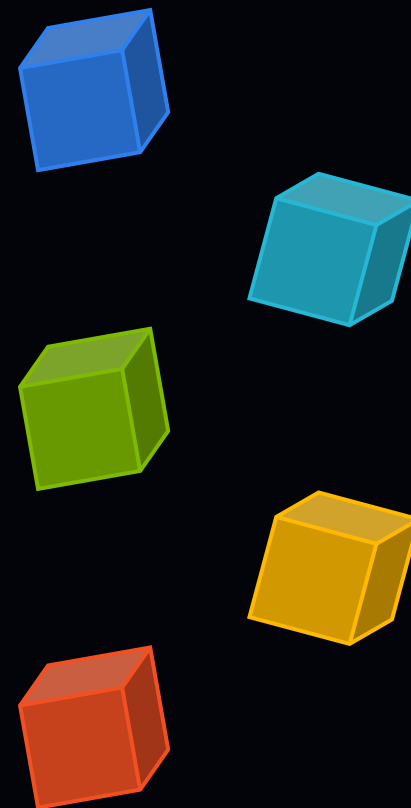


Data and feedback flow continuously.

Q & A

**It lacked context,
a definition of quality,
and an improvement loop.**

Now it has all three.



BUILD → GROUND → EVALUATE → OBSERVE → OPTIMISE → REPEAT